

Open Problems and Resolved Conjectures for *Lights Out* Nullity

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Abstract

Let $d(n)$ be the nullity over $\mathbb{F}_2$ of the bounded $n \times n$ *Lights Out* grid. Earlier notes in this directory mixed genuine open problems with statements subsequently proved in other manuscripts, standard Fibonacci-polynomial identities, computational observations, and several false parameterizations. This document is the canonical status list as of September 2026.

The surviving problems concern prime-power stabilization at unknown base-2 Wieferich primes, prime “mad” numbers, the global $4/5$ maximum-nullity bound and its record holders, independent impossible-nullity seeds, and the Most Clicks Problem beyond nullity two. We also record the exact fixed-odd-part theorem that supersedes the old b -factoring notebook, the computer-assisted proof that 34 is impossible, and explicit counterexamples to the retired formulations.

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1 Definitions and status conventions

Let

$$F_0(x) = 0, \quad F_1(x) = 1, \quad F_n(x) = xF_{n-1}(x) + F_{n-2}(x)$$

in $\mathbb{F}_2[x]$. Sutner's formula is

$$d(n) = \deg \gcd(F_{n+1}(x), F_{n+1}(x+1)) \quad (1)$$

[Sut00].

For a graph G , the Most Clicks Problem asks for the largest, over solvable configurations, of the minimum number of clicks needed to solve the configuration. We write $\text{MCP}(n)$ for the answer on the $n \times n$ grid.

The status descriptions below distinguish:

- *published theorem*: available in the cited literature;
- *repository theorem*: proved in another manuscript in this repository, but not necessarily published;
- *computer-assisted repository theorem*: reduced to a finite exact computation whose code and certificate are in the repository;
- *open*: no proof or counterexample is known;
- *retired*: proved, superseded, malformed, or false, and therefore no longer stated as a conjecture.

2 The fixed-odd-part theorem

This theorem disposes of the old b -specific polynomial conjectures and the meta-conjecture about the recurrence in k .

Theorem 2.1 (Fixed odd part). *Let b be odd, $k \geq 1$, $s = k - 1$, and $\epsilon_b = [3 \mid b]$. Define*

$$C_{b,k}(x) = \gcd(F_{b2^s}(x), F_{b2^s}(x+1)).$$

Then

$$C_{b,k}(x) = [x(x+1)]^{\epsilon_b(2^s-1)} C_{b,1}(x)^{2^s}. \quad (2)$$

Consequently

$$d(b2^{k-1} - 1) = 2^s d(b-1) + 2\epsilon_b(2^s - 1), \quad (3)$$

and

$$d(b2^k - 1) = 2d(b2^{k-1} - 1) + 2[3 \mid b]. \quad (4)$$

Proof. Hunziker, Machiavelo, and Park prove

$$F_{b2^s}(x) = x^{2^s-1} F_b(x)^{2^s}$$

[HMP04]; the same identity has recently been revisited in the classification of powerful Fibonacci polynomials [BJL⁺27]. Substitute $x+1$ for x .

For odd b , F_b is the square of a square-free polynomial. The factor $x+1$ divides $F_b(x)$ exactly when $3 \mid b$, and then its multiplicity is exactly two; symmetrically, x divides $F_b(x+1)$ exactly in the same case. Neither x nor $x+1$ divides $C_{b,1}$. Taking the gcd of the two displayed factorizations gives (2). Degrees give (3), and comparing s with $s+1$ gives (4). The corresponding nullity recurrence is also proved in [Boy22b]. ■

Corollary 2.2. *For every $k \geq 1$,*

$$d(31 \cdot 2^{k-1} - 1) = 5 \cdot 2^{k+1}, \quad d(33 \cdot 2^{k-1} - 1) = 11 \cdot 2^k - 2.$$

Proof. The exact evaluations $d(30) = 20$ and $d(32) = 20$ give the result from (3). Equivalently, the Kloosterman calculation in `tex/upper_bound/upper_bound.tex` gives $G(31) = 20$ and $G(33) = 22$. ■

Remark 2.3. *The explicit formulas formerly conjectured for $b = 1, 3, 5, 7, 9$ are all instances of the published Fibonacci factorization used above. The “zero once, zero always” conjecture and the claimed recurrence $d(b, k+1) \in \{2d(b, k), 2d(b, k) + 2\}$ are both immediate consequences of (4).*

3 Resolved and retired claims

Former claim	Status	Current source or correction
Rule-90 and rule-150 coefficient recurrences	Published theorem / direct substitution	Sutner’s coefficient formula and the substitution $x \mapsto x + 1$ give the two cellular-automaton recurrences [Sut00]. They are expository identities, not conjectures.
$F_{b2^s} = x^{2^s-1}F_b^{2^s}$ and the b -factoring iteration	Published theorem	Theorem 2.1; see [HMP04, BJL ⁺ 27].
The $b = 31$ and $b = 33$ nullity formulas	Repository theorem	Corollary 2.2 and <code>tex/upper_bound/upper_bound.tex</code> .
Eventual constancy of $d(a^k - 1)$ for fixed odd a	Repository theorem	The finite-support truncation theorem in <code>tex/multiprime_stabilization/multiprime_stabilization.tex</code> .
26 is impossible, and impossibility propagates by $A \mapsto 2A + 2$	Repository theorem	<code>tex/finite_fields/finite_fields.tex</code> .
34 is impossible	Computer-assisted repository theorem	<code>tex/finite_witness_bounds/finite_witness_bounds.tex</code> ; the independent exact rerun checks all 32 767 candidates and reproduces, for the 16 100 newly computed pairs, SHA-256 64cc5c41bf1bf197c833f695e3aaf3308b4154b358667f1a4f79
All nullity-12 grids have size $340k - 256$	False	$d(594) = 12$, and $594 = 340 \cdot 2 - 86$. The corrected necessary classes are $n \equiv 84 \pmod{170}$.
The displayed higher-nullity MCP quadratics in the old summary	False as written	They used the subsequence index in place of the multiplier $q = (n+1)/a$; for example the old formula predicts $\text{MCP}(14) = 48$, while the exact value is 123.
Every varying polynomial gcd is a geometric series with a power-of-two number of terms	False	For $b = 15$, $C_{15,1} = 1 + x^2 + x^4$ has three terms, while $C_{15,2} = x + x^2 + x^5 + x^6 + x^9 + x^{10}$ has six and is not a geometric series.

4 Prime-power stabilization

For a prime $p > 5$, put

$$c_p = v_p(2^{\text{ord}_p(2)} - 1).$$

Every non-Wieferich prime has $c_p = 1$.

Conjecture 4.1 (Wieferich endpoint). *For every prime $p > 5$ and every $k \geq 1$,*

$$d(p^k - 1) = d(p - 1).$$

The statement is a theorem for every non-Wieferich prime. More generally,

$$d(p^k - 1) = d(p^{c_p} - 1) \quad (k \geq c_p).$$

The two known base-2 Wieferich primes, 1093 and 3511, both have $c_p = 2$, and exact endpoint computations give

$$d(p^2 - 1) = d(p - 1) = 0.$$

Thus Conjecture 4.1 holds for every currently known prime and, in particular, for every $p < 2^{64}$.

The companion note `tex/prime_power_endpoint/prime_power_endpoint.tex` also gives analytic all-level proofs for both known Wieferich primes. More generally, for every prime $p > 5$, either $3 \mid \text{ord}_p(2)$ or $v_2(\text{ord}_p(2)) = 2$ implies $d(p^k - 1) = 0$ for all $k \geq 1$, without any restriction on Wieferich depth. The two cases include 3511 and 1093, respectively. The remaining unresolved primes must therefore be Wieferich and satisfy $3 \nmid \text{ord}_p(2)$ and $v_2(\text{ord}_p(2)) \neq 2$.

The minimal unresolved assertion is the endpoint lemma: if θ has order p^{c_p} and

$$1 + \theta^a + \theta^{-a} + \theta^b + \theta^{-b} = 0,$$

then $p^{c_p-1} \mid a, b$. See `tex/five_term_plateau/five_term_plateau.tex` for the equivalent polynomial, cyclotomic, coding, dynamical, elliptic, and group-theoretic forms.

5 Prime mad numbers

For a prime $p > 3$, let

$$R_p = \{\zeta + \zeta^{-1} : \text{ord}(\zeta) = p\} \subseteq \overline{\mathbb{F}}_2.$$

Proposition 5.1. *For every prime $p > 3$,*

$$d(p - 1) = 2 |R_p \cap (R_p + 1)|.$$

Thus $d(p - 1) > 0$ exactly when there are nontrivial p -th roots t, s satisfying

$$1 + t + t^{-1} + s + s^{-1} = 0.$$

This is the bounded-grid form of the prime Button Madness problem studied through Dickson polynomials in [BCCH18].

Problem 5.2 (Arithmetic characterization). *Characterize the primes p for which $R_p \cap (R_p + 1) \neq \emptyset$.*

Conjecture 5.3 (Prime infinitude). *There are infinitely many primes p for which $d(p - 1) > 0$.*

Every Mersenne prime greater than 7 and every Fermat prime greater than 3 has this property; exact Kloosterman formulas appear in `tex/five_term_plateau/five_term_plateau.tex`. Brouwer and Blokhuis prove infinitely many mad *integers*, but that does not prove Conjecture 5.3, since infinitude of the relevant prime subfamily remains open [Bro]. The necessary condition

$$\rho_2(p) = \min\{r > 0 : 2^r \equiv \pm 1 \pmod{p}\} \leq \sqrt{p}$$

is not sufficient: $\rho_2(241) = 12 < \sqrt{241}$ but $d(240) = 0$.

6 Maximum nullity

Conjecture 6.1 (The 4/5 bound). *For every $n \geq 1$,*

$$5d(n) \leq 4(n+1),$$

with equality if and only if $n = 5 \cdot 2^k - 1$ for some $k \geq 0$.

The equality family itself is proved:

$$d(5 \cdot 2^k - 1) = 4 \cdot 2^k.$$

The manuscript `tex/upper_bound/upper_bound.tex` reduces the global bound to a multiplicative-subgroup point count on an elliptic curve and proves it:

- when the odd part of $n+1$ is $2^r \pm 1$;
- under an explicit Weil-type criterion;
- for every odd base $b \leq 10^5$ by exact computation.

In all these regimes $b = 5$ is the unique equality case. No published or repository proof of the uniform bound for every b is known. The strongest general published comparison remains Sarkar's theorem $d(n) = n$ if and only if $n = 4$ [Sar96].

Define the positive record-holder set

$$\mathcal{M} = \{n \geq 1 : d(n) > 0 \text{ and } d(n) > d(m) \text{ for every } m < n\}.$$

Conjecture 6.2 (Record holders).

$$\mathcal{M} = \{5 \cdot 2^{k-1} - 1 : k \geq 1\} \cup \{31 \cdot 2^{k-1} - 1 : k \geq 1\} \cup \{33 \cdot 2^{k-1} - 1 : k \geq 2\}.$$

This is verified for all $n \leq 100000$. The exclusion of $33 \cdot 2^0 - 1 = 32$ is necessary because $d(32) = 20$ only ties the earlier record at $n = 30$.

7 Impossible nullities

Let

$$I = \{A \in 2\mathbb{N} : A \neq d(n) \text{ for every } n\}.$$

If $A \in I$, call it a *seed* when it is not obtained from a smaller member of I by iterating $A \mapsto 2A + 2$.

The repository proves that 26 and 34 are seeds. The doubling recurrence then gives the two infinite impossible families

$$2^t(26 + 2) - 2 \quad \text{and} \quad 2^t(34 + 2) - 2 \quad (t \geq 0).$$

The finite-witness theorem in `tex/finite_witness_bounds/finite_witness_bounds.tex` further proves that membership in the image of d is decidable by a finite computation for each prescribed value.

Conjecture 7.1 (Infinitely many seeds). *The set I contains infinitely many seed values.*

8 Small nullities and the Most Clicks Problem

The two-adic formula (3), together with the ranks of the small translation-invariant Fibonacci factors, gives the following global necessary conditions.

Theorem 8.1 (Small-nullity congruences).

$$\begin{aligned} d(n) = 4 &\implies n \equiv 4 \pmod{10}, \\ d(n) = 6 &\implies n \equiv 11 \pmod{24}, \\ d(n) = 8 &\implies n \equiv 9 \pmod{20} \text{ or } n \equiv 16 \pmod{34}, \\ d(n) = 10 &\implies n \equiv 29 \pmod{60}, \\ d(n) = 12 &\implies n \equiv 84 \pmod{170}. \end{aligned}$$

Proof. Write $n + 1 = 2^s b$, with b odd, and put $D = d(b - 1)$. For even boards $b - 1$, one has $4 \mid D$. Solving

$$d(n) = 2^s D + 2[3 \mid b](2^s - 1)$$

for the five target values gives the possible pairs (s, D) .

If $D = 4$, the square-free half of the common Fibonacci factor is $x^2 + x + 1$, of Fibonacci rank 5, so $5 \mid b$. If $D = 8$, the corresponding translation-invariant factor is $x^4 + x + 1$, of rank 17, so $17 \mid b$. If $D = 12$, the only viable degree-six factor has rank 85. The two alternative rank-63 factors occur together and would force nullity at least 24, so $85 \mid b$. Substitution gives the displayed congruences. The exact rank calculation is recorded in `tex/finite_witness_bounds/finite_witness_bounds.tex` and the supporting nullity code. ■

For nullity two, the repository proves that every grid has size $6k - 1$ and

$$\text{MCP}(6k - 1) = 26k^2 - 12k + 1;$$

see `tex/nullity2/nullity2.tex` and [Boy22a]. No higher-nullity family has been proved.

The following corrected fits use the natural multiplier $q = (n + 1)/a$, not the subsequence index that appeared in the old summary.

Conjecture 8.2 (Empirical MCP fits). *For each row below, if $n = aq - 1$ with q odd and $d(n)$ has the stated value, then $\text{MCP}(n) = P(q)$.*

$d(n)$	a	$P(q)$
4	5	$17q^2 - 10q$
6	12	$88q^2 - 24q + 1$
8	10	$60q^2 - 20q - 3$
8	17	$161q^2 - 34q - 3$
10	30	$506q^2 - 60q - 3$

The rows agree with respectively 19, 6, 7, 6, and 4 exact values in `code/serialization/min_clicks_answers.txt`, but no proof is known. The former nullity-12 fit is not retained: even $\text{MCP}(84)$ is known only within the interval $3579 \leq \text{MCP}(84) \leq 3591$, and the old source used inconsistent coefficients.

Problem 8.3 (Higher-nullity MCP). *Determine $\text{MCP}(n)$ on any infinite family of grids with nullity at least four. More generally, determine whether the MCP is polynomial or quasi-polynomial in the multiplier q on a fixed nullity family.*

9 Computational evidence and maintenance

The nullity data in `code/serialization/b159257.txt` contains every $d(n)$ for $1 \leq n \leq 100000$. It verifies:

- Conjecture 6.1, with the stated equality cases, throughout that range;
- Conjecture 6.2 throughout that range;
- all congruences in Theorem 8.1;
- the necessity of both nullity-8 classes (the first omitted old case is $d(69) = 8$);
- both mod-340 subprogressions of the single nullity-12 class $n \equiv 84 \pmod{170}$ (the first omitted old case is $d(594) = 12$, with $594 \equiv 254 \pmod{340}$).

The old standalone b -factoring, maximum-nullity, and rule-90/rule-150 notes have been removed from the conjectures directory. Their valid content is either Theorem 2.1, a cited published identity, or material in the dedicated manuscripts named above. Generated PDFs and unused figures were removed with them so that this source remains the only status document in the directory.

References

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